

Module 6: Linux Server - Manage Basic Networking & Security

Section 1: Networking Basics

49. Use `ifconfig` or `ip` to view and configure network interfaces.

These commands are used to display or modify the configuration of the system's network interfaces.

- **ip addr show:** The modern standard to view IP addresses and interface details.
- **ifconfig:** The legacy command (requires net-tools) to view interface details.
- **To assign an IP temporarily:** `sudo ip addr add 192.168.1.50/24 dev eth0`

50. Use `ping` to test network connectivity.

`ping` sends ICMP Echo Request packets to a destination to check if it is reachable.

- **Command:** `ping <IP_or_Domain>` (e.g., `ping 8.8.8.8`)
- **Stop:** Press `Ctrl + C` to stop the infinite loop and see statistics.

56. How to Set Static IP in Linux?

The method varies by distribution, but the most common modern way (on Ubuntu/Debian) is using **Netplan**:

1. Edit the config file: `sudo nano /etc/netplan/01-netcfg.yaml`
2. Configure the interface:

YAML

```
network:
  version: 2
  ethernets:
    eth0:
      dhcp4: no
      addresses: [192.168.1.100/24]
      gateway4: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

3. Apply changes: `sudo netplan apply`
(On RHEL/CentOS, you edit files in `/etc/sysconfig/network-scripts/` or use `nmtui`).

Section 2: Firewall Management

51. Understand basic firewall configuration using FIREWALL-CMD.

firewall-cmd is the command-line interface for **firewalld**, used in RHEL/CentOS/Fedora.

- **Check status:** `sudo firewall-cmd --state`
- **List all settings:** `sudo firewall-cmd --list-all`
- **Reload firewall:** `sudo firewall-cmd --reload`

52. Add SSH services in firewall.

To allow SSH traffic through the firewall permanently:

1. **Command:** `sudo firewall-cmd --permanent --add-service=ssh`
2. **Apply:** `sudo firewall-cmd --reload`

53. Graphically manage the firewall.

For users who prefer a GUI, the tool is called **firewall-config**.

- **Install:** `sudo apt install firewall-config` (Debian) or `sudo dnf install firewall-config` (RHEL).
- **Run:** Type `firewall-config` in the terminal or search for "Firewall" in the application menu.
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Section 3: Advanced Security

55. What is SELinux Security?

SELinux (Security-Enhanced Linux) is a kernel security module that provides a mechanism for supporting access control security policies.

- **Concept:** It uses **Mandatory Access Control (MAC)**. Unlike standard permissions where a user can grant access to their files, SELinux follows a central policy that defines what every process can do.
- **Modes:**
 - **Enforcing:** Security policy is enforced.
 - **Permissive:** Policy is not enforced, but actions are logged.
 - **Disabled:** SELinux is completely off.
- **Check Status:** `sestatus`